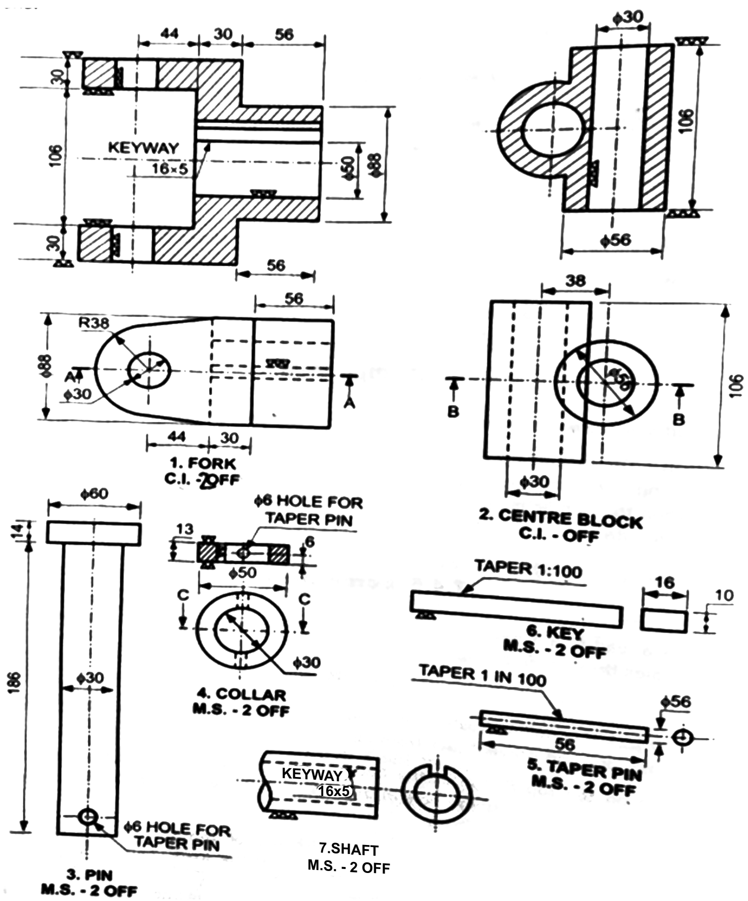

*Dimension units may be taken in mm

*Assume suitable dimensions as require if not specified or not visible clearly in given fig.

Q.27 Draw the views of following by taking appropriate dimensions.

a) Leaf spring

b) Connecting Rod

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Roll No. ....

4th Semester/ Auto

Subject: Automobile Engineering Drawing

Time : 3 Hrs. M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

Q.1 The basic dimension for which allowable maximum and minimum limits are decided in called

a) Basic size b) Actual size

c) Large size d) Small size

Q.2 How many forks are there is universal coupling

a) 1 b) 2

c) 3 d) 4

Q.3 Material used for piston is

a) Copper b) Aluminium

c) Sulpur d) Magnisium

Q.4 Connecting rod connects piston with

a) Clutch b) Flywheel

c) Cam shaft d) Crankshaft

Q.5 The number of minimum valves in a cylinder are

a) 1 b) 2

c) 3 d) 4

Q.6 Wheel cylinder is the part of

a) Engine b) Transmission

c) Brake d) Suspension

Q.7 When the tolerance distribution is only on one side of the basic size, it is known as

a) Unilateral tolerance b) Bilateral tolerance

c) Compound tolerance d) Geometric tolerance

Q.8 Gudgeon pin is also known as

a) Crank pin b) Screw pin

c) Piston pin d) None of above

Q.9 Which of the following is not a part of the transmission system?

a) Clutch b) Flywheel

c) Gear box d) Axles

Q.10 Which of the following is not a part of IC engine?

a) Piston b) Crank shaft

c) Radiator d) Flywheel

SECTION-B

Note: Objective/ Completion type questions. Attempt any five questions. (5x2=10)

Q.11 Define basic dimension.

Q.12 Define limits.

Q.13 Define rise in case of cam

Q.14 What is the function of crankshaft?

Q.15 What is the function of Hook's joint?

Q.16 Name two types of cam.

SECTION-C

Note: Draw free hand sketch of any two. (2x10=20)

Q.17 Ball bearing.

Q.18 Side valve mechanism.

Q.23 Wheel cylinder.

SECTION-D

Note: Attempt any two questions out of four questions. (30x2=60)

Q.24 Draw a profile of cam having knife edge follower to give following motion

a) Outstroke during 60° of cam rotation

b) Dwell for the next 30° of cam rotation

c) Return stroke during next 60° of cam rotation

d) Dwell for remaining 210° of cam rotation.

The stroke of the follower is 50 mm and min. Radius for the cam is 60mm. The follower moves with uniform velocity during both strokes. The axis of cam passes through axis of follower.

Q.25 Draw the profile of involute teeth by approximate method having 24 teeth, module pitch 12 mm pressure angle of 22°. Draw minimum three teeth.

Q.26 Figure shows detailed drawing of a universal coupling. Draw the following orthographic views.

a) Front view full sectional

b) Top view